ME 652 Project Final Report

Multibody Dynamic Modeling of a Tilt-quadrotor eVTOL

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1.0 Introduction to the eVTOLs

Electric Vertical Takeoff and Landing (eVTOL) aircraft can overcome traditional runway limitations and adapt to confined spaces and complex terrains, showing broad application prospects in building urban transportation networks and accelerating the low-carbon transition of the aviation industry. In this context, system modeling and flight control design for eVTOLs are of great strategic significance. Currently eVTOLs are primarily categorized into three typical configurations.

The first configuration is the tail-sitter aircraft, which utilizes propellers on the wings for vertical takeoff and tilts its entire body horizontally to transition from hovering to forward flight. Because both flight phases share the same propulsion system. The second configuration is the tiltrotor aircraft. Unlike the tail-sitter, its fuselage remains horizontal while only the rotors tilt. The rotors face upward to provide lift during vertical operations and gradually tilt forward to provide forward thrust as speed increases. The third configuration is the dual-system aircraft, which features two independent propulsion systems: a multirotor system for vertical takeoff and hovering, and a pusher motor for horizontal cruising.

Figure 1: Conceptual Applications of eVTOL (Air Taxi) [1][2]

Based on this technical background, this project focuses on the research of a tilt-quadrotor eVTOL. The aircraft consists of three main components: the fuselage, the rotors, and the tilt mechanism. The rotor section features four composite propellers arranged in an X-

shaped layout, combined with a tilt mechanism capable of omnidirectional movement, allowing the aircraft to accomplish complex flight tasks.

2.0 Problem Statement and Motivation

The dynamic behavior of tilt rotor eVTOL systems is inherently complex due to the strong coupling between rotors, tilt-ring and drone body. This leads to nonlinear force distributions and varying load conditions across different flight modes, making simplified analytical models insufficient for accurate prediction. In addition, a multibody dynamics approach is required to properly capture the motion and force interactions within tilt-rotor systems [3]. Therefore, there is a need to develop a reliable simulation framework to evaluate joint forces and assess structural performance under realistic operating conditions.

With the growing demand for urban air mobility, eVTOL aircraft have emerged as a promising transportation solution. However, ensuring structural safety and performance under complex flight conditions remains a significant challenge. Previous studies have shown that accurate modeling of tiltrotor dynamics is essential for capturing nonlinear behavior and system interactions. [4] This project aims to integrate multibody dynamics simulation (ADAMS) with structural analysis (FEA) to provide a systematic method for predicting joint loads and validating component designs, ultimately contributing to safer and more efficient eVTOL development.

3.0 The CAD Design of the Tiltrotor eVTOL

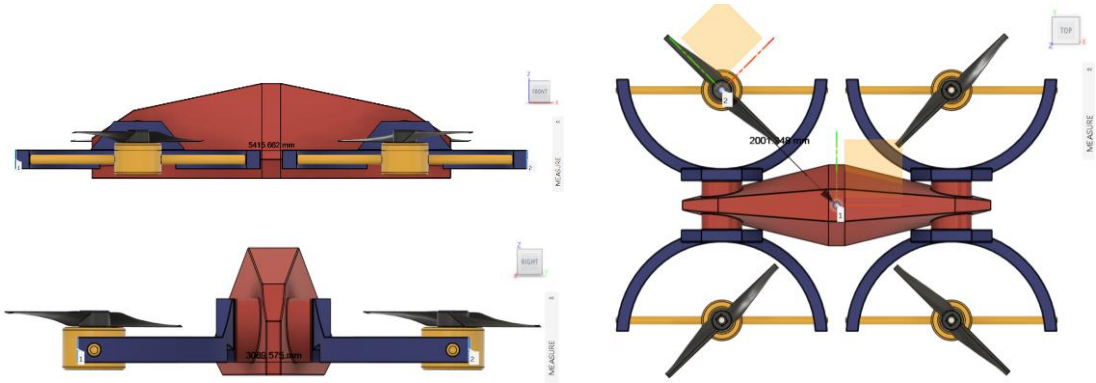


Figure 2: Fusion 360 CAD Modeling

The Fusion 360 CAD assembly shown in Figure 2 represents the proposed drone model developed in this project. The design includes the drone body, tilt-ring mechanisms, motor assemblies, and propellers in a quadcopter configuration, with overall dimensions of approximately 5416 mm (length) and 3000 mm (width). In subsequent stages, the model will be integrated with ADAMS and MATLAB/Simulink for dynamic analysis and FEA-driven iterative design optimization.

4.0 Dynamic Modeling of the Tiltrotor eVTOL

4.1 Coordinate Systems and Transformations

To establish the flight model, three standard right-handed orthogonal coordinate systems are defined [5]: the Ground coordinate ($O_g x_g y_g z_g$), the Body ($O_b x_b y_b z_b$) origin at center of mass), and the Airflow ($O_a x_a y_a z_a$) coordinate systems.

The transformation from the ground to the body frame is defined by the yaw (ψ), pitch (θ), and roll (ϕ) angles [5]:

$$R_g^b(\phi,\theta,\psi) = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ -s\phi s\theta c\psi - c\phi s\psi & -s\phi s\theta s\psi + c\phi c\psi & s\phi c\theta \\ c\phi s\theta c\psi + s\phi s\psi & c\phi s\theta s\psi - s\phi c\psi & c\phi c\theta \end{bmatrix}$$

Here $c \triangleq \cos(\cdot)$, $s \triangleq \sin(\cdot)$. Position transforms can be written as:

$$[x_g, y_g, z_g]^T = R_b^g [x_b, y_b, z_b]^T$$

The body-to-airflow transformation uses angle of attack (α) and sideslip (β)[5]:

$$R_b^a(\alpha,\beta) = \begin{bmatrix} \cos\beta\cos\alpha & \sin\beta & \cos\beta\sin\alpha \\ -\sin\beta\cos\alpha & \cos\beta & -\sin\beta\sin\alpha \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$$

Airspeed V_a relates to ground speed V_g and wind speed: $V_a = V_g - V_w$.

Body velocity is denoted as $V_b = [u, v, w]^T$ and ground velocity is denoted as $V = [v_x, v_y, v_z]^T$。

4.2 Aircraft Forces and Moments

Gravity: In the body frame, gravity force F_g^b and moment M_g^b are:

$$F_g^b = R_g^b \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} = \begin{bmatrix} -mg\sin\theta \\ mg\cos\theta\sin\phi \\ mg\cos\theta\cos\phi \end{bmatrix} \quad M_g^b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Rotor Dynamics: For the quadrotor eVTOL, individual rotor lift F and anti-torque Q can be simplified as: $F = K_T\omega^2, Q = K_Q\omega^2$[6]. The total lift F_R and local moments T_x, T_y, T_z (before tilting) are:

$$\begin{bmatrix} F_R \\ T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} K_T & K_T & K_T & K_T \\ \frac{\sqrt{2}}{2}dK_T & \frac{\sqrt{2}}{2}dK_T & -\frac{\sqrt{2}}{2}dK_T & -\frac{\sqrt{2}}{2}dK_T \\ \frac{\sqrt{2}}{2}dK_T & -\frac{\sqrt{2}}{2}dK_T & -\frac{\sqrt{2}}{2}dK_T & \frac{\sqrt{2}}{2}dK_T \\ K_Q & -K_Q & K_Q & -K_Q \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix}$$

where d is the distance from the rotor center to the center of mass. The mechanical tilt mechanism's rotation matrix R_t (pitch η, roll γ) is:

$$R_t(\gamma,\eta) = R_x(\gamma)R_y(\eta) = \begin{bmatrix} \cos\eta & 0 & \sin\eta \\ \sin\eta\sin\gamma & \cos\gamma & -\cos\eta\sin\gamma \\ -\sin\eta\cos\gamma & \sin\gamma & \cos\eta\cos\gamma \end{bmatrix}$$

The total rotor force F_r^b projected into the body frame is:

$$F_r^b = R_t(\gamma,\eta) \begin{bmatrix} 0 \\ 0 \\ -F_R \end{bmatrix} = \begin{bmatrix} -F_R\sin\eta \\ F_R\cos\eta\sin\gamma \\ -F_R\cos\eta\cos\gamma \end{bmatrix}$$

Let J_R be individual rotor inertia, and $\Omega_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$ be the residual angular velocity. The body-frame rotor angular momentum H_r^b is:

$$H_r^b = R_t(\gamma,\eta) \begin{bmatrix} 0 \\ 0 \\ J_R\Omega_r \end{bmatrix} = J_R\Omega_r \begin{bmatrix} \sin\eta \\ -\cos\eta\sin\gamma \\ \cos\eta\cos\gamma \end{bmatrix}$$

The gyroscopic moment M_{gyro}^b due to body rotation $\omega = [p, q, r]^T$ is calculated as:

$$M_{gyro}^b = -\omega \times H_r^b$$

The total rotor moment M_r^b combines the tilted aerodynamic control moments and the gyroscopic moment:

$$M_r^b = R_l(\gamma,\eta) \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} + M_{gyro}^b$$

Aerodynamics: Modeled primarily as drag, the aerodynamic force F_a^b acts against the velocity vector V_b with a lumped coefficient k_d:

$$F_a^b = -k_d V_a V_b = -k_d \sqrt{u^2 + v^2 + w^2} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

4.3 Dynamics and Kinematics Equations

Treating the aircraft as a rigid body, the Newton-Euler dynamic equations are:

$$\begin{cases} \dot{V} = \dfrac{F^b}{m} \\ \dot{\omega} = I^{-1}(\omega \times I\omega + M^b) \end{cases} \qquad V = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \ \omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad I = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

$$F^b = F_a^b + F_r^b + F_g^b \qquad M^b = M_r^b$$

The 6-DOF kinematics equations include the position navigation equations:

$$V = R_g^b(\phi, \theta, \psi) V^b$$

And the attitude rate equations:

$$\begin{cases} \dot{\phi} = p + (r\cos(\phi) + q\sin(\phi))\tan(\theta) \\ \dot{\theta} = q\cos(\theta) - r\sin(\phi) \\ \dot{\psi} = \dfrac{1}{\cos(\theta)}\big(r\cos(\phi) + q\sin(\phi)\big) \end{cases}$$

The constructed Simulink mode is shown below:

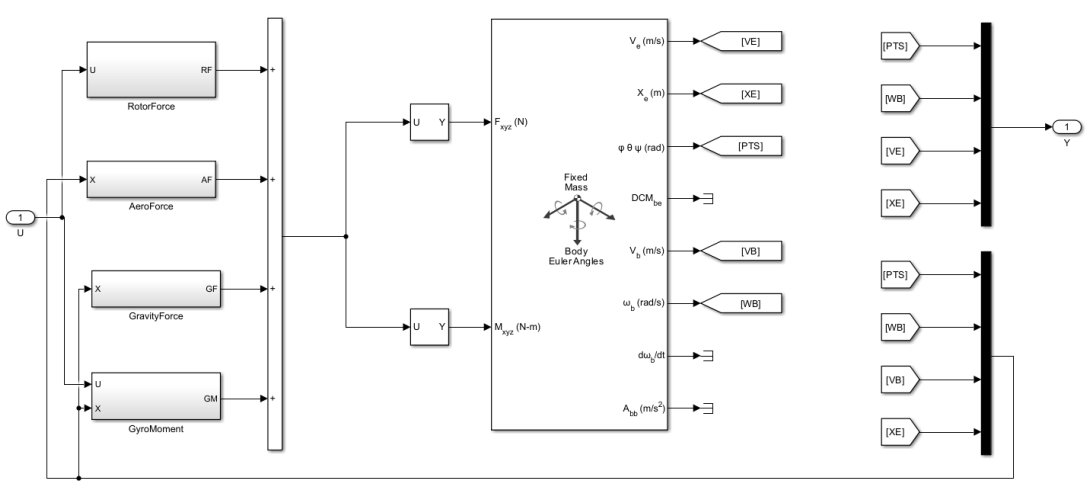
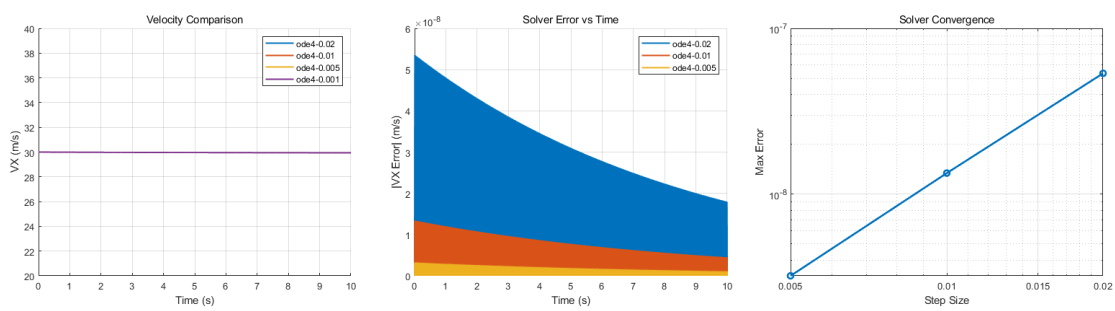


Figure 3 eVTOL dynamic model in Simulink

5.0 eVTOL Model Verification and Flight Characteristics Analysis

5.1 Forward dynamics verification

To verify the accuracy of the established model, the project adopts a forward kinematics verification approach. Since real flight test data is unavailable in a laboratory setting, the rotor speeds and tilt angles for a steady level flight at 30 m/s were derived through flight dynamics principles. These parameters were then input into the model to observe whether the kinematic response matches the predefined flight state. Furthermore, the solver type and step size were varied to assess the model's convergence consistency. The verification results below demonstrate that the model's response is perfectly aligned with the preset flight conditions. Moreover, changes in solver configuration do not affect the convergence of the results, thereby confirming the robustness and accuracy of the model.



(a) (b) (c)

Figure 4: Forward dynamic verification results. (a): Velocity comparison across different step sizes; (b): Absolute solver error over time; (c): Convergence rate showing the relationship between step size and maximum error.

5.2 Flight Characteristics Analysis

Rather than the forward dynamics problem, we are more concerned with the inverse dynamics problem — determining the required rotor speeds and tilt angles for a given flight state. The equations of motion and dynamics derived in the previous section are variable-coefficient, nonlinear equations which can only be solved using numerical methods. Trimming [5] refers to finding the steady-state operating point of the system where the total external force and moment are balanced, and state variables like angle of attack, speed, and control surface deflections remain constant (i.e., their derivatives are zero).

The project selects the three types of steady flight states for analysis: level fight, planar diagonal fight and spatial diagonal flight to test its omni-directional mobility and direction vectors are shown below. The trim operation uses the Simulink SQP algorithm, setting state constraints $(\phi, \theta, \psi,\ u, v, w, p, q, r) = 0$. The trim curves obtained for the 3 flight modes are shown in figures below:

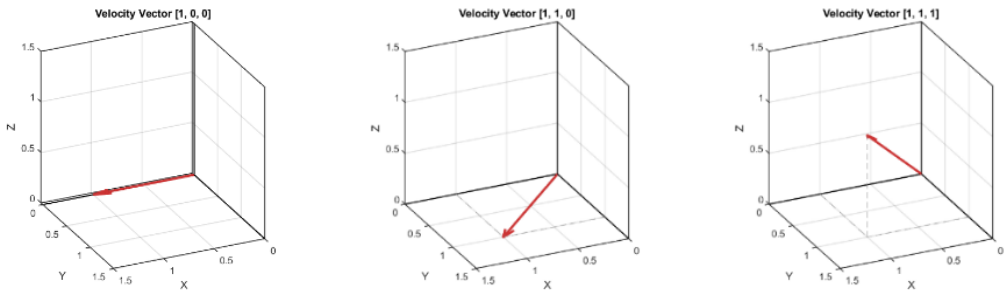


Figure 5: Different flight states. Left: level flight; Middle: planar diagonal fight; Right: spatial diagonal flight

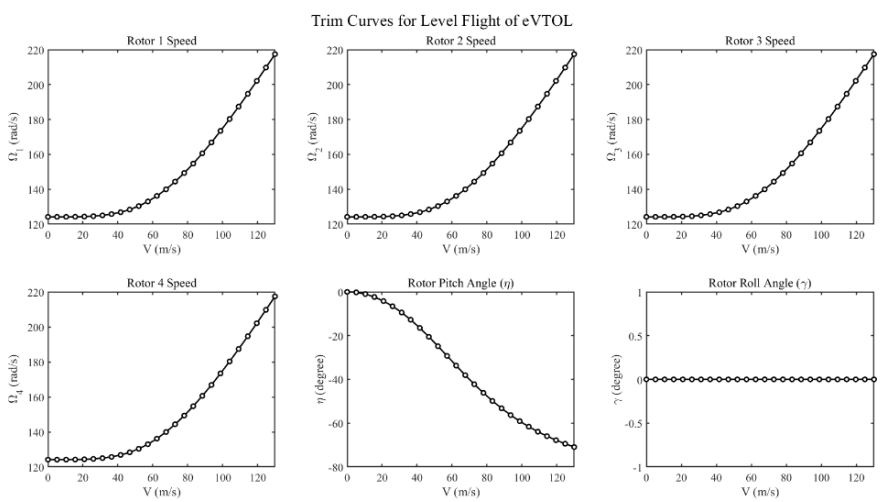


Figure 6 Trim curves for level fight

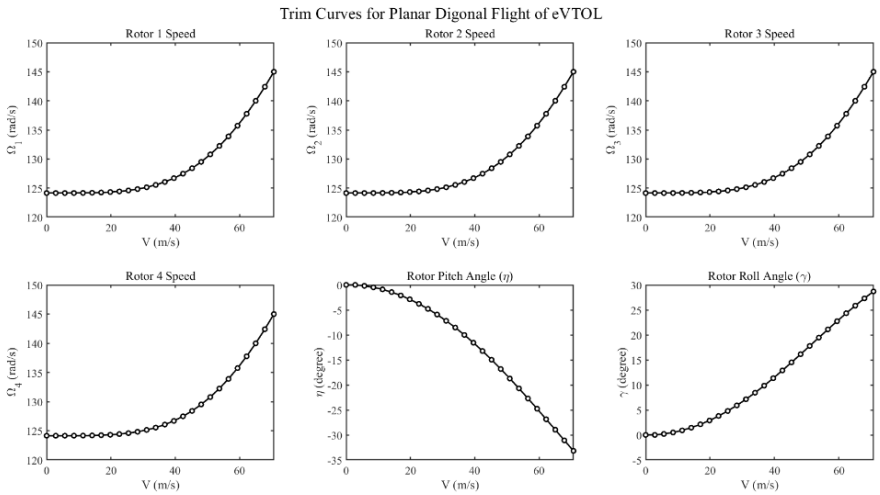


Figure 7 Trim curves for planar diagonal flight

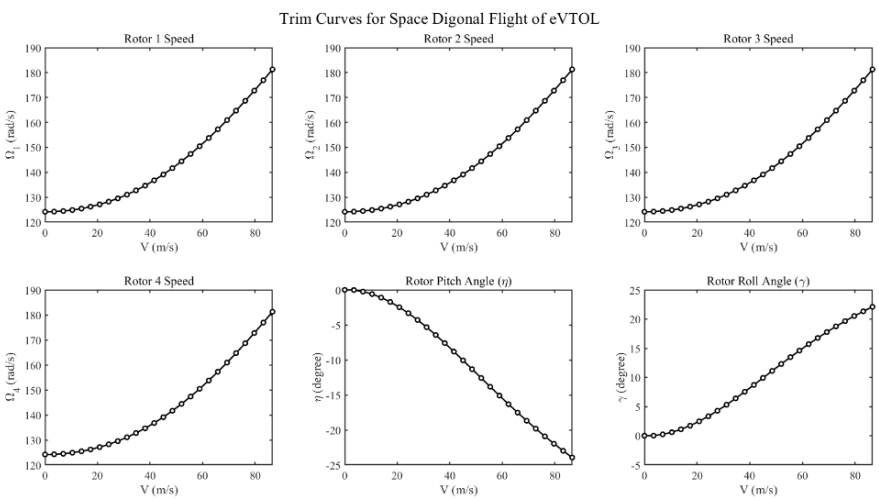


Figure 8 Trim curves for spatial diagonal flight

In the pure level flight mode, forward motion is achieved entirely by pitching the rotors forward; as the flight velocity V increases, the rotor pitch angle η changes steadily to overcome drag, while the roll angle γ remains perfectly at zero. For the planar and space diagonal flights, the aircraft needs to travel along off-axis trajectories, so both tilt mechanisms are engaged. This is shown by the simultaneous use of a negative pitch angle η and a positive roll angle γ to balance the aerodynamic forces in 3D space.

Across all three flight modes, the rotational speeds of all four rotors ($\omega_1\ to\ \omega_4$) are essentially identical at any given velocity. This indicates that during steady flight, the

aircraft does not rely on differential thrust to maintain its attitude or heading. Instead, it uses the mechanical tilt angles to vector the thrust for directional control, while the motors simply scale up their RPM to provide the required total thrust. This control allocation strategy is highly beneficial, as it keeps the power load evenly distributed.

6.0 Control Design

6.1 Reduced model control

Instead of applying the full nonlinear 6-DOF dynamics, a reduced linear model is developed for the initial control design. In this approach, the controller output is defined as the desired translational acceleration, which is then mapped to the corresponding rotor thrust, pitch tilt angle η, and roll tilt angle γ. Based on this, the following equations can be formulated, where x represents the state vector and u represents the control input in the form of commanded accelerations.

$$x = \begin{bmatrix} x \\ \dot{x} \\ \mathrm{y} \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} u = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} where\ \dot{x_1} = x_2,\ \dot{x_2} = u_x,\ \dot{x_3} = x_4,\ \dot{x_4} = u_y,\ \dot{x_5} = x_6,\ \dot{x_6} = u_z$$

Then for the reduced model, the state space representation in $\dot{x} = Ax + Bu, y = Cx$ becomes:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

In addition, the reduced model is verified to be both controllable and observable. The matrix, $M_c = [B \quad AB \quad A^2B \quad A^3B \quad A^4B \quad A^5B]$ have rank of 6, and $M_o = [C \quad CA \quad CA^2 \quad CA^3 \quad CA^4 \quad CA^5]^T$ also have full rank. Therefore, the reduced system is suitable for controller design.

Ideally, the control input $u = [u_x \quad u_y \quad u_z]^T$ will map to the eVTOL plant input where $u = [\omega_1 \quad \omega_2 \quad \omega_3 \quad \omega_4 \quad \eta_1 \quad \eta_2 \quad \eta_3 \quad \eta_4 \quad \gamma_1 \quad \gamma_2 \quad \gamma_3 \quad \gamma_4]^T$ by following allocation/transformation equations:

$$T = \rho A \omega_i^2 R^2 C_T, \;\; \omega_i = \sqrt{\frac{m\sqrt{u_x^2 + u_y^2 + (u_z + g)^2}}{4\rho A R^2 C_T}}, \;\; \eta_i = \arctan\frac{u_x}{u_z + g}, \gamma_i = \arctan\frac{u_y}{u_z + g}$$

After the allocation equations are defined, state-feedback control can be introduced for position tracking. The tracking error is defined as $e = x - x_{ref}$, where x_{ref} is the desired position commands.

For the feedback gain matrix, take the control input law $u = -K(x - x_{ref})$, where gain K is obtained using the Linear Quadratic Regulator (LQR) method that minimizes the quadratic cost function:

$$J = \int_0^\infty (x^T \mathrm{Q} x + u^T R u) dt$$

Then choose the matrix Q and R where weight the importance of state or control to get the gain K (for example let Q be diag[50 5 50 5 100 10] and R be diag[1 1 1], solving the ARE, can get following gain K).

$$K = \begin{bmatrix} 7.0711 & 4.3752 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7.0711 & 4.3752 & 0 & 0 \\ 0 & 0 & 0 & 0 & 10 & 5.4772 \end{bmatrix}$$

As a result, after takeoff, and the target position is set to x at 100 and y at 50, the simulation below does show a result that is not ideal. Although the reduced model can drive eVTOL toward the desired region, the velocity does not converge to zero. Showing that the eVTOL still retains residual translational motion after reaching the target region (blue curve).

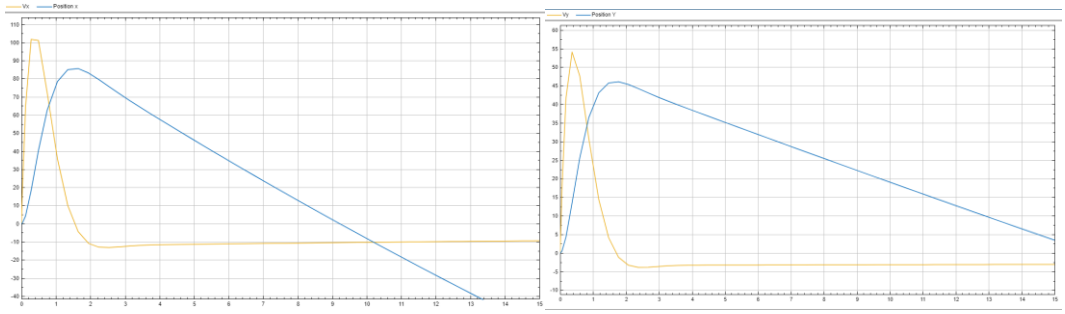


Figure 9 Position & Velocity Behavior

In addition, the p, q, and r are small in 10^{-15}, indicating that the eVTOL stays nearly level and experiences negligible rotational motion. To improve the result, the velocity v_x, v_y, v_z are feed back into the allocator with gains to account for the difference between the ideal reduced model and the nonlinear eVTOL plant.

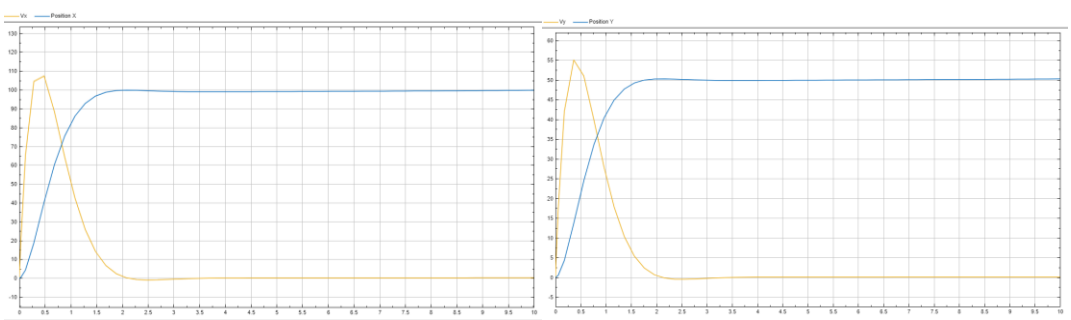


Figure 10 Position & Velocity Behavior with Gain

As shown in the plots above, the feedback weighted gains in the allocator improves the overall tracking response of the plant, able to correct the final velocity to zero, and gives the result that the eVTOL stop at the desired final location.

And the general reduced system build is shown in following figure, where the left side presents the simplified state-space model with feedback gain, while the right side shows the allocator, which converts the accelerations into the desired rotor inputs:

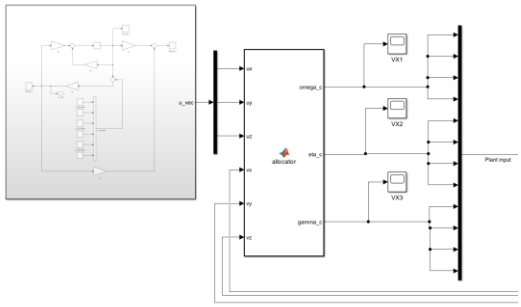


Figure 11 General reduced Model

Generally, the reduced model can convert the position commands in x, y and z directions and the simulation results show that the vehicle is able to move toward the target locations with small p, q and r states (remain level). However, the reduced model neglects important nonlinear effects present in the eVTOL plant, including aerodynamic drag, actuator dynamics, and coupling between translational and rotational motion. As a result, residual velocity mismatch can still appear near the final position.

Continue to the linear model control. One solution is to linearize using trim data from some motion modes shown in previous section, for example the level motion, diagonal motion, and space motion. Recall the following trim input set in previous section, the system has six control inputs. The first four inputs represent the squared rotor speeds, ω^2, of the 4 rotors. Assume that all rotors operate at the same speed in the trim condition, these four inputs are equal for each trim point. The 5th input is the motor pitch angle, η, and the sixth input is the motor roll angle,γ.

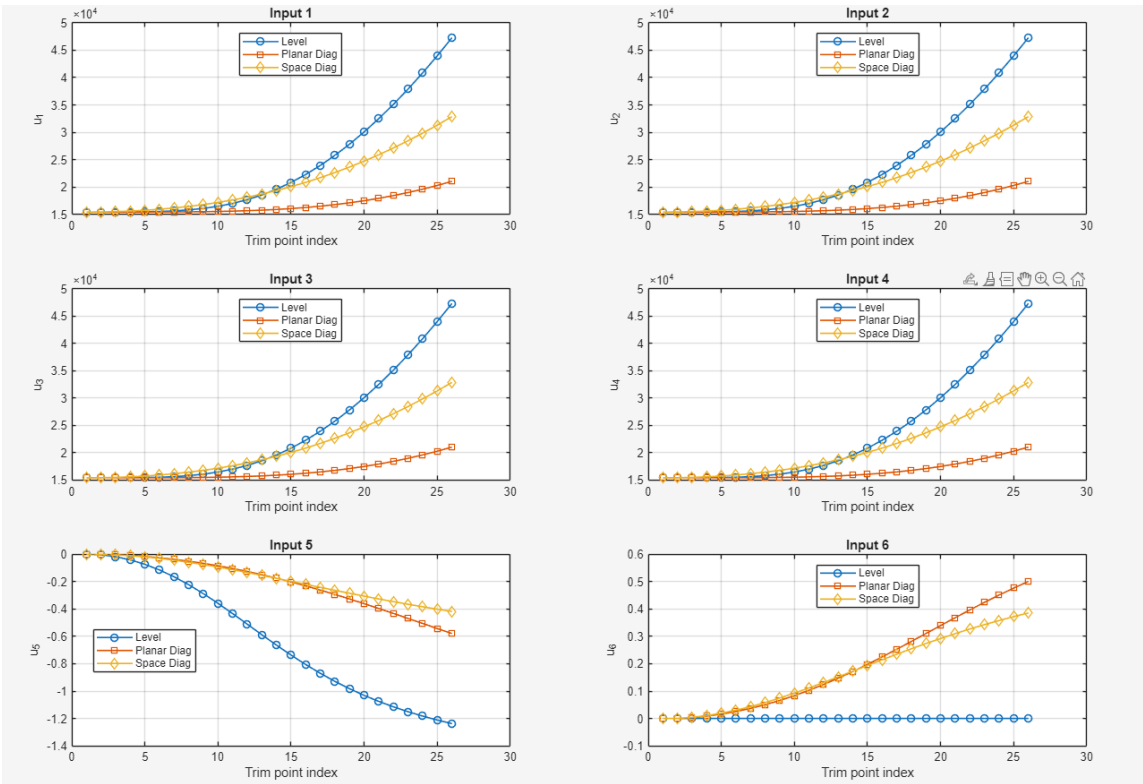


Figure 12 Trim Data used in Linearization

Therefore, by using trim data, total thrust, and the arm and motor angles in different flight modes, linearization using selected trim points can be efficient for the nonlinear eVTOL plant. Take u_0, assume at steady operating point, from the plant, which has following input and output:

$$u = [\omega_1 \quad \omega_2 \quad \omega_3 \quad \omega_4 \quad \eta_1 \quad \eta_2 \quad \eta_3 \quad \eta_4 \quad \gamma_1 \quad \gamma_2 \quad \gamma_3 \quad \gamma_4]^T$$

$$x = [\varphi \quad \Theta \quad \psi \quad p \quad q \quad r \quad v_x \quad v_y \quad v_z \quad x \quad y \quad z]^T$$

Then simulate the nonlinear plant near the trim condition, that runs open loop using the trim input u_0. After that, use input perturbation, and output measurement in Simulink to perform the linearization, using 1st order Taylor expansion:

$$\dot{x} = f(x,u), \qquad y = h(x,u), \qquad \delta x = x - x_0, \qquad \delta u = u - u_0$$

$$\delta\dot{x} = A\delta x + B\delta u, \qquad A = \frac{df}{dx}|x_0, u_0, \qquad B = \frac{df}{du}|x_0, u_0$$

$$\delta y = C\delta x + D\delta u, \qquad C = \frac{dh}{dx}|x_0, u_0, \qquad D = \frac{dh}{du}|x_0, u_0$$

Then calculate the Jacobian matrices from the block diagram. After this, use same approach as the reduced model that gives similar results. As a result, the plant can be approximated by a local linear state-space model.

6.2 Dynamic Feedback Linearization Control

Considering the limitations of the previous control system design, we have developed a new system by adopting the methodology from a published paper [7]. The tilt-quadrotor eVTOL is a fully actuated system. It possesses six independent control inputs: the four squared rotor speeds, denoted as $W = [\omega_1^2, \omega_2^2, \omega_3^2, \omega_4^2]^T$, and the two tilting mechanism rates, denoted as $\dot{\eta}_t = [\dot{\gamma}, \dot{\eta}]^T$.

6.2.1 System Reformulation

To apply feedback linearization, we first rewrite the translational and rotational dynamic equations in a compact matrix form. The total rotor force F_r^b and moment M_r^b are linear with respect to the squared rotor speeds W:

$$F_r^b = F_m(\gamma, \eta)W$$
$$M_r^b = T_m(\gamma, \eta)W$$

where $F_m = \frac{\partial F_r^b}{\partial W} \in \mathbb{R}^{3\times4}$ and $T_m = \frac{\partial M_p^b}{\partial W} \in \mathbb{R}^{3\times4}$ are Jacobian matrices dependent on the tilt angles.

The Newton-Euler equations from Section 4.3 can be aggregated as:

$$\begin{bmatrix} \ddot{P} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} g_v \\ c \end{bmatrix} + \begin{bmatrix} \frac{1}{m} R_b^g & 0_{3\times3} \\ 0_{3\times3} & I^{-1} \end{bmatrix} \begin{bmatrix} F_m & 0_{3\times2} \\ T_m & 0_{3\times2} \end{bmatrix} \begin{bmatrix} W \\ \dot{\eta}_t \end{bmatrix}$$

where $g_v = [0,0,g]^T$ represents the gravity vector in the ground frame, and $c = -I^{-1}(\omega \times (I\omega))$ encompasses the gyroscopic and Coriolis terms. Let the transformation matrix be $J_R = \mathrm{diag}\left(\frac{1}{m} R_b^g, I^{-1}\right)$. The system mapping matrix $J = J_R \begin{bmatrix} 0 & 0 \\ T_m & 0 \end{bmatrix} \in \mathbb{R}^{6\times6}$ is singular due to its two rightmost null columns. Consequently, the system cannot be statically feedback linearized.

6.2.2 Dynamic Feedback Linearization

To circumvent this singularity, the system can be linearized at a higher differential order. By differentiating the acceleration equations with respect to time, the control input derivatives (e.g., tilt angle velocities) are exposed:

$$\begin{bmatrix} \ddot{P} \\ \ddot{\omega} \end{bmatrix} = \begin{bmatrix} \frac{1}{m}\dot{R}_b^g F_r^b \\ \dot{c} \end{bmatrix} + J_R \frac{d}{dt}\left(\begin{bmatrix} F_m \\ T_m \end{bmatrix} W\right)$$

Applying the chain rule to the terms involving F_m and T_m:

$$\frac{d}{dt}(F_m W) = F_m \dot{W} + \left(\frac{\partial F_m}{\partial \gamma} W\right)\dot{\gamma} + \left(\frac{\partial F_m}{\partial \eta} W\right)\dot{\eta}$$

$$\frac{d}{dt}(T_m W) = T_m \dot{W} + \left(\frac{\partial T_m}{\partial \gamma} W\right)\dot{\gamma} + \left(\frac{\partial T_m}{\partial \eta} W\right)\dot{\eta}$$

This allows us to define the extended Jacobian matrix $J^* \in \mathbb{R}^{6\times 6}$:

$$J^* = J_R \begin{bmatrix} F_m & \frac{\partial F_m W}{\partial \gamma} & \frac{\partial F_m}{\partial \eta} W \\ T_m & \frac{\partial T_m}{\partial \gamma} W & \frac{\partial T_m}{\partial \eta} W \end{bmatrix}$$

The fully differentiated system is thus governed by:

$$\begin{bmatrix} \ddot{P} \\ \ddot{\omega} \end{bmatrix} = J^* \begin{bmatrix} \dot{W} \\ \dot{\eta}_t \end{bmatrix} + \begin{bmatrix} \frac{1}{m}\ddot{R}_b^g F_r^b \\ \dot{c} \end{bmatrix}$$

Provided that the condition $\det(J^*) \neq 0$ is satisfied (which holds true as long as the total thrust is non-zero and the tilt angles remain within mechanical limits, i.e., $|\gamma|, |\eta| < \pi/2$), the extended Jacobian is invertible. The dynamic feedback linearization control law can then be formulated as:

$$[\dot{W}] = (J^*)^{-1}\left(\begin{bmatrix} \ddot{P}_r \\ \dot{\omega}_r \end{bmatrix} - \begin{bmatrix} \frac{1}{m}\dot{R}_b^g F_r^b \\ \dot{c} \end{bmatrix}\right)$$

where $\ddot{P}_r$ and $\dot{\omega}_r$ are the virtual reference inputs for translational jerk and angular acceleration rate, respectively. The actual actuator inputs, W and η_l, are obtained by numerically integrating the outputs $\dot{W}$ and $\dot{\eta}_l$.

6.2.3 Trajectory Tracking Controller

To ensure the output tracking error converges exponentially to zero, proportional-derivative controllers are designed to track a desired trajectory defined by position $P_d(t)$, orientation $R_d(t)$, and their higher-order derivatives.

The linear trajectory tracking controller for the translational jerk reference is given by:

$$\dddot{P}_r = \dddot{P}_d + K_{p_1}(\ddot{P}_d - \ddot{P}) + K_{p_2}(\dot{P}_d - \dot{P}) + K_{p_3}(P_d - P)$$

Similarly, the attitude tracking controller is formulated as:

$$\dddot{\omega}_r = \dddot{\omega}_d + K_{\omega_1}(\dot{\omega}_d - \dot{\omega}) + K_{\omega_2}(\omega_d - \omega) + K_{\omega_3} e_R$$

where K_{p_i} and K_{ω_i} $(i = 1,2,3)$ are positive-definite diagonal gain matrices placed to assure proper pole placement. The orientation error $e_R \in \mathbb{R}^3$ maps the $SO(3)$ rotation manifold to a vector using the vee operator $[\,.\,]_V$:

$$e_R = \frac{1}{2}\left[R_d^T R_b^g - (R_b^g)^T R_d\right]_V$$

Similar to the original paper, we performed numerical simulations of the control system. We tested two scenarios: a pure level motion with constant body orientation and a complex combined motion that includes both position and attitude motion.

Translational Motion: The UAV demonstrates pure translational movement along the X and Y axes while maintaining a constant null body orientation $(\phi, \theta, \psi = 0)$. The mechanism's tilt angles (η, γ) compensate for the lateral acceleration, allowing the body to remain level throughout the flight.

Coupled Motion: The results show the platform's ability to simultaneously track changes in both position (p_x) and body attitude (θ). Specifically, the body pitch can be set to desired values during translational motion, a maneuver that is physically impossible for standard underactuated quadrotors.

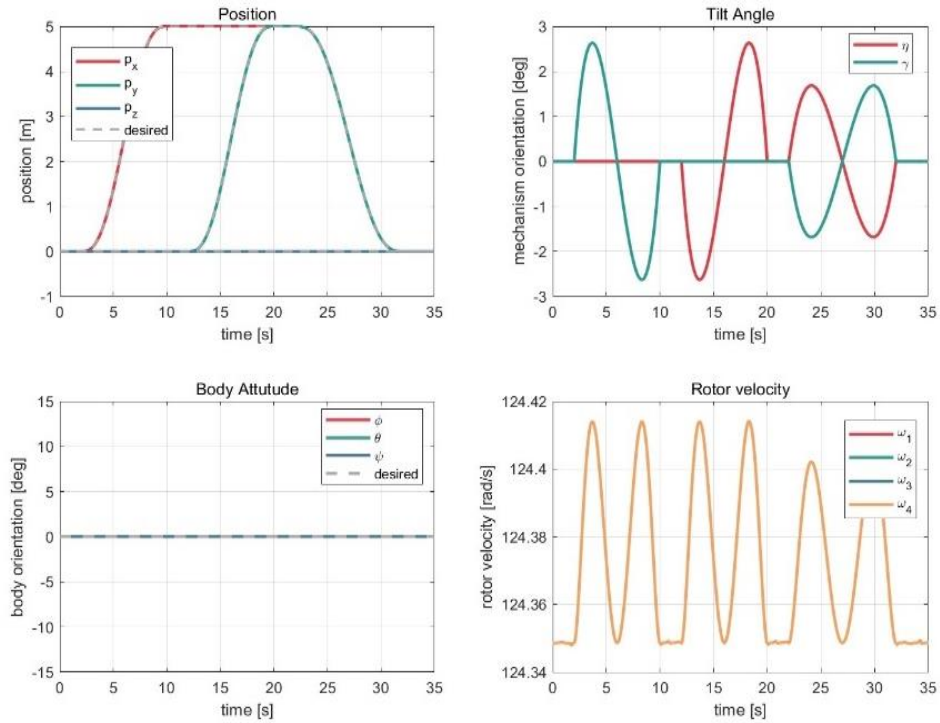


Figure 13 Results of translational motion

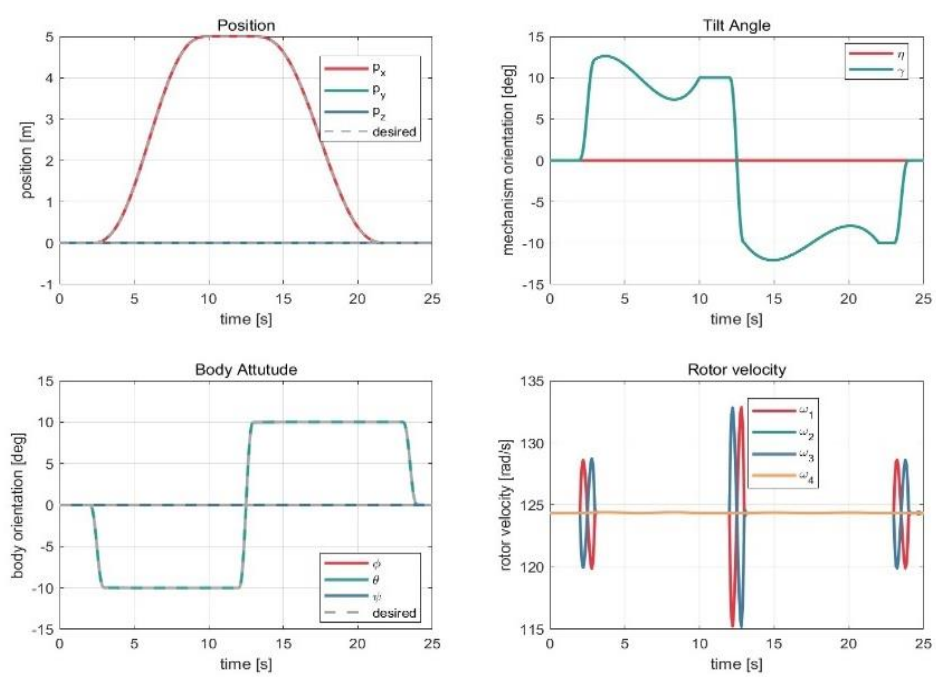


Figure 14 Results of the coupled motion

Generally, the control design demonstrates that both the linearization approach and the dynamic nonlinear approach are useful for the eVTOL system. The trim-based linearization method approximates the nonlinear dynamics near selected points, which is used in initial controller design. However, its validity is inherently limited to the vicinity of the chosen trim points, and it does not account for significant nonlinear effects.

For comparison, the dynamic feedback linearization method is derived directly from the full nonlinear model. It addresses the singularity issue through reformulation and differentiation of the system outputs, thereby enabling the controller to exploit the complete capability of the actuators for simultaneous position and attitude control. Simulation results confirm that eVTOL achieves stable translational motion, maintains level attitude when required, and successfully performs coupled position-attitude maneuvers. Overall, the development of control shows that the linearized model serves as a practical tool for preliminary design, while the dynamic nonlinear approach delivers superior robustness and more comprehensive control performance for the eVTOL system.

7.0 Analysis of eVTOL by ADAMS

In Adams, the trim data can also be used to export as spline functions. First build a speed sequence from zero to the maximum speed of each motion mode, using $V_{\max}$ and number of trim points. Then use spline acceleration, to convert to time sequence that can avoid sharp changes in acceleration:

$$t = T_{max}\frac{2}{\pi}\arcsin\left(\sqrt{\frac{V}{V_{max}}}\right)$$

In addition, the rotor thrust $T(t)$, the vertical thrust, and reaction torque $\tau(t)$ can be calculated using:

$$T(t) = C_T\rho AR^2\omega^2(t),$$
$$T_{vertical}(t) = C_T\rho AR^2\omega^2(t)\cos\eta(t)$$
$$\tau(t) = C_\tau\rho AR^3\omega^2(t)$$

Then, import the converted spline functions into ADAMS:

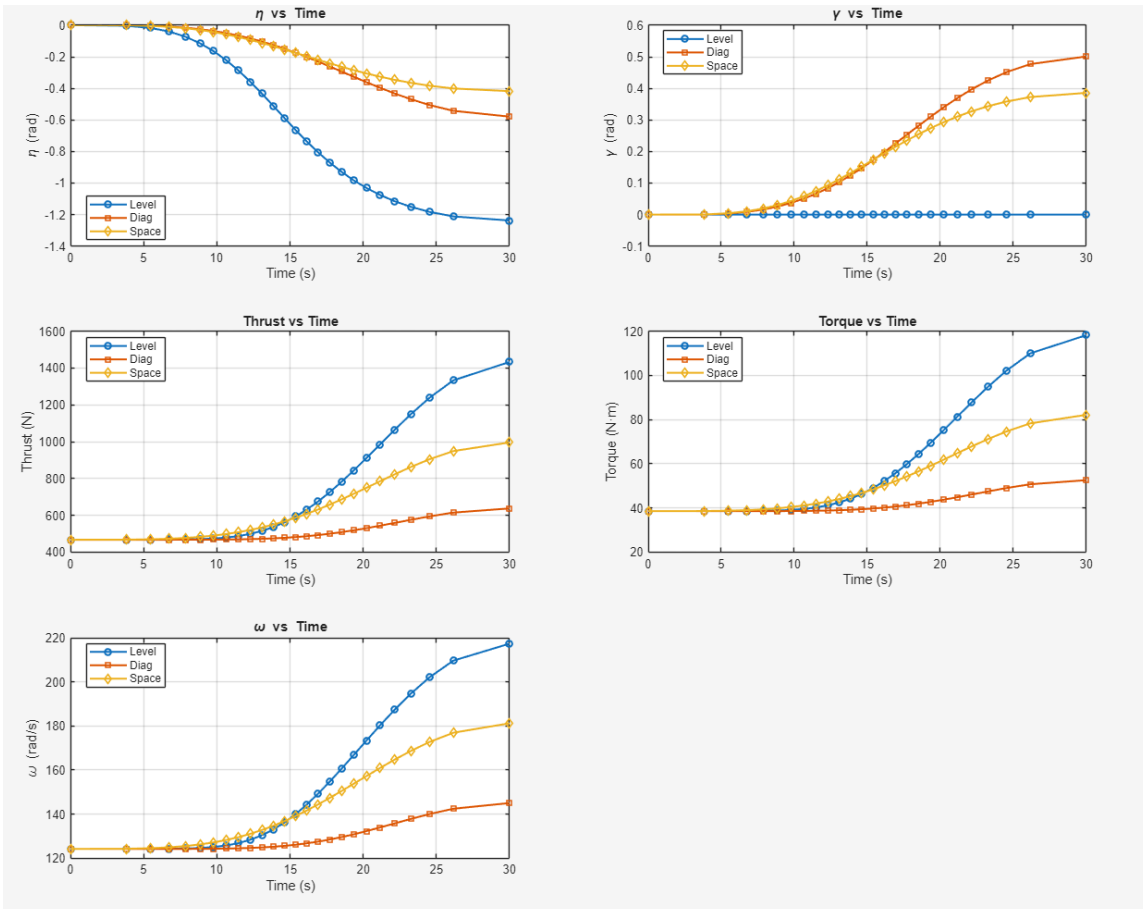


Figure 15 Spline Function used for Adams

For ADAMS simulation, after importing the CAD model, three rotational joints are assigned as motion drives for prescribed actuator motions, one external force is applied for the rotor thrust and one external torque is applied for the reaction torque. To define the joint motions and loads, the Spline with Akima fitting method is used. Therefore, the multibody model can reproduce the required actuator motion and loading conditions for the subsequent dynamic analysis.

Figure 16 shows the proposed CAD model and its implementation in ADAMS. The main components include the drone casing, tilt ring, blades, and motor assembly, with their material properties, mass, volume, and moments of inertia listed in Table 1. To improve simulation efficiency, only one quadrant is modeled in ADAMS, while the remaining three are suppressed, allowing effective evaluation of joint forces and dynamic behavior.

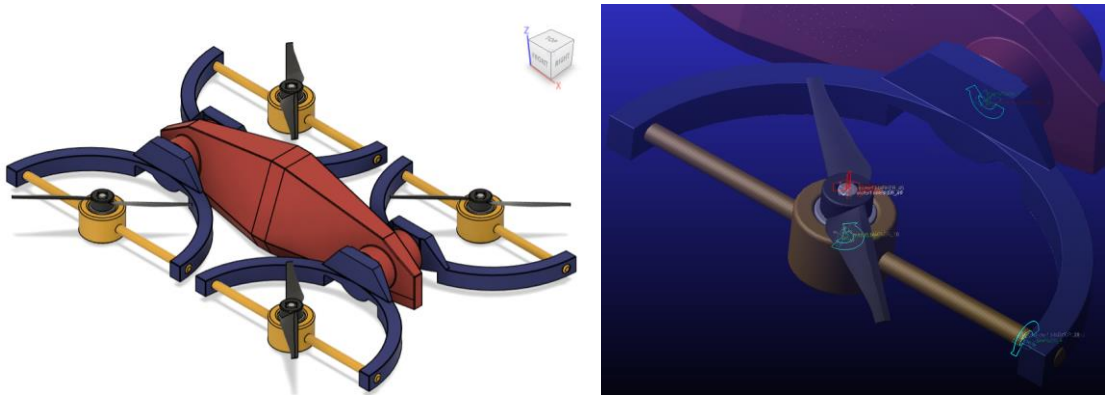


Figure 16: Drone CAD Model and ADAMS Simulation Setup

Components	Material	Mass (kg)	Volume (m^3)	Ixx ($kg \cdot mm^2$)	Iyy ($kg \cdot mm^2$)	Izz ($kg \cdot mm^2$)
Drone Casing	CFRP	47.913	3.351×10^{-2}	6.736×10^{9}	5.868×10^{10}	5.716×10^{10}
Tilt Ring	CFRP	11.974	8.373×10^{-3}	2.244×10^{9}	7.784×10^{9}	9.747×10^{9}
Blades (72")	CFRP Nylon	1.150	7.67×10^{-4}	1.60×10^{5}	1.60×10^{5}	3.20×10^{5}
Motor Assembly	CFRP Aluminum	13.015	7.331×10^{-3}	2.249×10^{9}	2.918×10^{9}	2.267×10^{9}

Table 1: Major Component Mass, Volume and Inertias

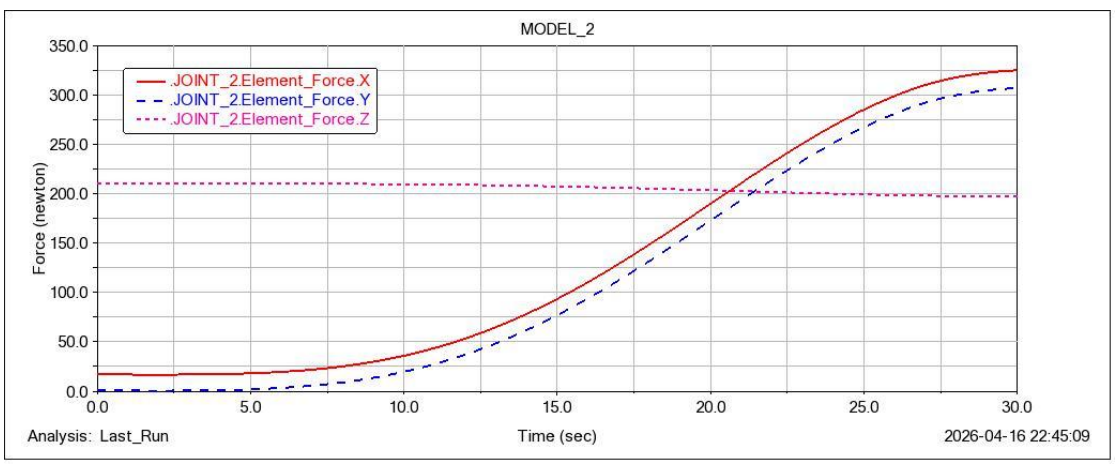


Figure 17: Joint Forces in Diagonal Flight Mode

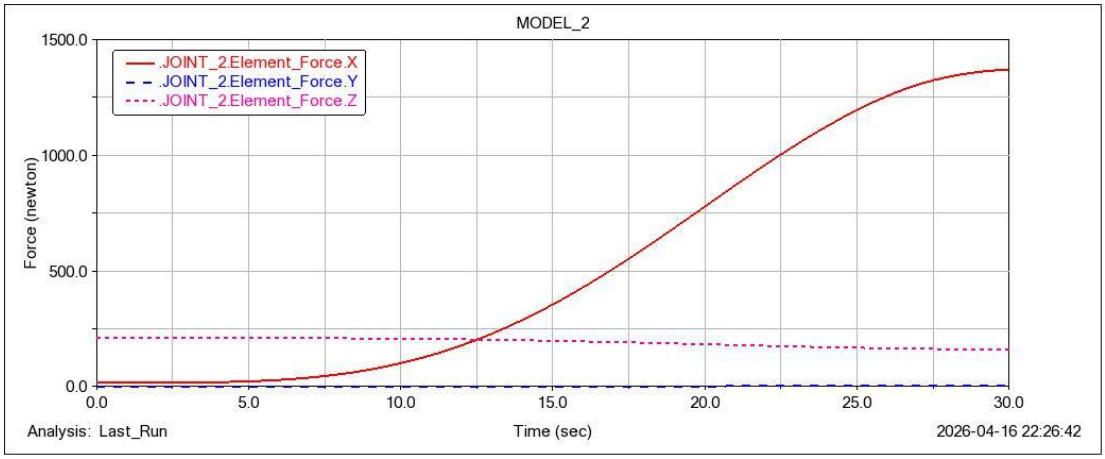


Figure 18: Joint Forces in Level Flight Mode

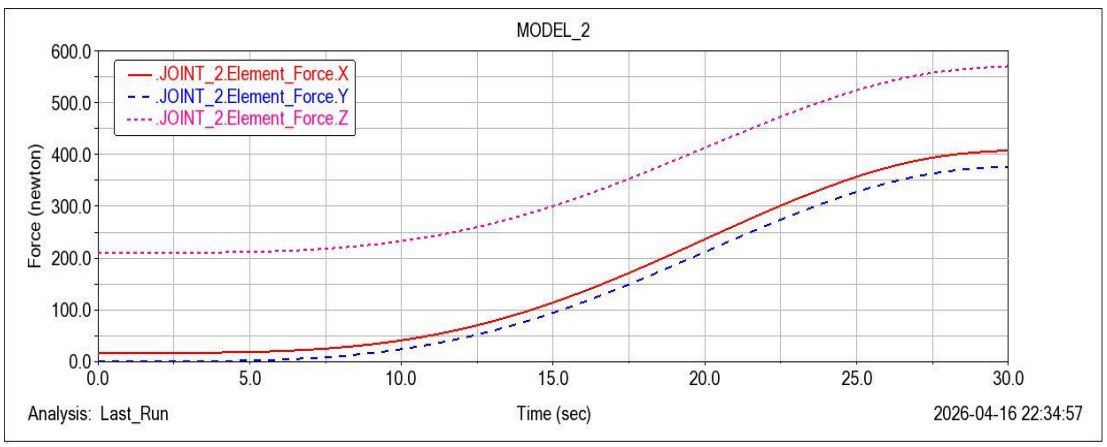


Figure 19: Joint Forces in Spatial Flight Mode

7.1 Finite Element Analysis

Based on the joint force results from the three cases (Figure 17,18 and 19), the maximum loading condition is selected to evaluate whether the proposed CAD design can withstand the applied loads. The level flight case is identified as the critical condition, with forces of $F_x = 1400\,\mathrm{N}$, $F_y = 0\,\mathrm{N}$, and $F_z = 250\,\mathrm{N}$. A static structural simulation is conducted in Fusion 360 with gravity applied, assuming CFRP material properties. The resulting safety factor, von Mises stress, and displacement distributions are presented in the following figures.

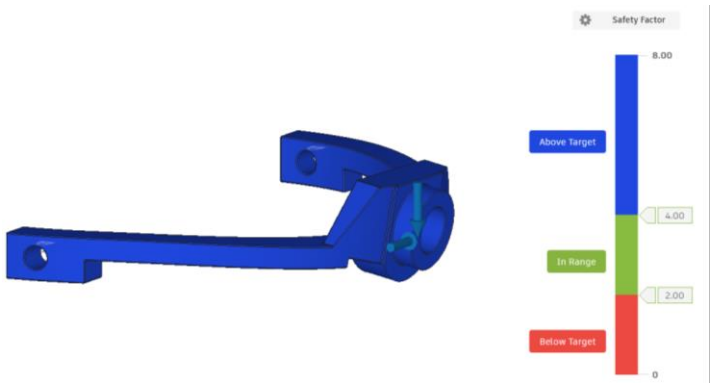


Figure 20: Safety Factor Distribution of the Tilt Arm

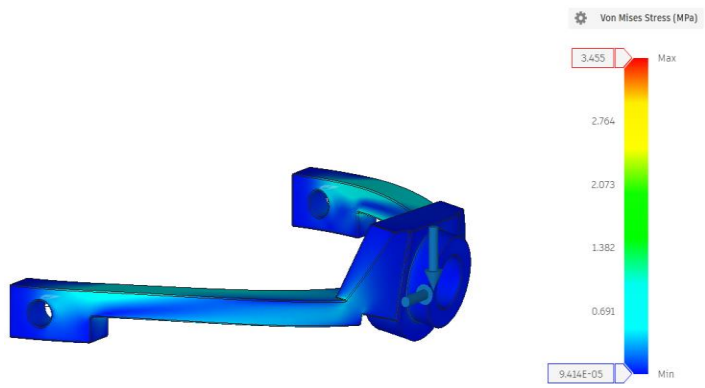


Figure 21: Von Mises Stress Distribution of the Tilt Arm

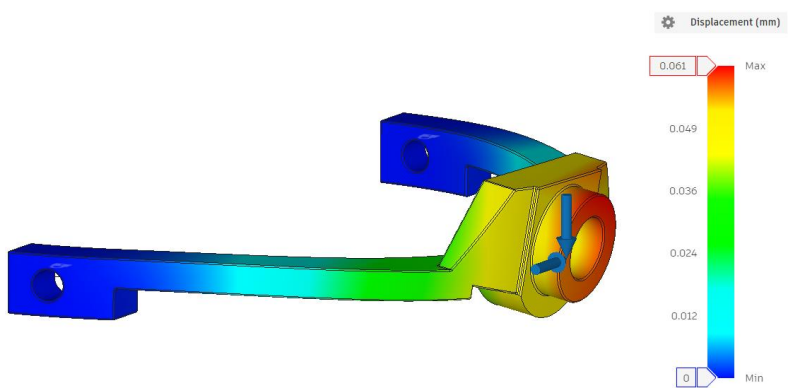


Figure 22: Displacement Distribution of the Tilt Arm

The results show that the tilt arm maintains strong structural performance under the maximum loading condition. As seen in Figure 20, the safety factor is well above the required threshold, indicating a conservative design. From Figure 21, stress concentrations appear near the joint and curved regions, but the overall stress level remains low for CFRP.

As shown in Figure 22, the maximum displacement is approximately 0.061 mm, which is minimal and does not affect structural stability.

Overall, the design is structurally safe, with potential for further weight optimization. However, the current geometry may not be fully manufacturable in practice. Future work will focus on refining the design with more detailed and realistic features, as well as applying a wider range of boundary conditions to perform FEA on additional components.

8.0 Conclusion

This project developed a multibody dynamic model of a tilt rotor eVTOL system integrated with structural analysis and simple control system design. The results show that the approach can capture system dynamics, predict joint forces and provide benchmarks for control inputs with good accuracy under different flight conditions. In addition to the reduced linear model control and dynamic non-linear control, they can both demonstrate the position control for the eVTOL system, while the nonlinear approach provides more robust performance. In forward dynamics analysis, we validate the model's accuracy by comparing its output with theoretical predictions. In Adams part, we used a simplified dynamic model calculated in MATLAB as a benchmark. By comparing the behavioral trends of the MATLAB results with the Adams output, we can determine whether the Adams simulation is physically consistent and reasonable. Future work will focus on incorporating more realistic boundary conditions in both FEA and fluid dynamics simulations to ensure the proposed components achieve good manufacturability while satisfying required force and torque constraints. Additionally, a scaled prototype will be developed and tested to further validate the proposed methodology, as well as the reliability and accuracy of the overall workflow. Besides, the CM of the motor and tilting arm is aligned with the fuselage CM plane to eliminate the dynamic effects of CM translation during rotation. We plan to investigate non-coplanar configurations in the next phase. Additionally, the tilting mechanism is currently modeled as a simplified skeleton. Future work will involve detailing the internal components to enable a more rigorous and accurate multi-body dynamics simulation. For the control section, additional flight modes as well as the implementation of a recursive design workflow that updates the physical properties directly from the latest CAD model can be made, for example, route control that has SoC

aware optimization, for better analysis and application on the eVTOL model. Although some design simplifications limit manufacturability, the framework provides a solid basis for future refinement and optimization.

9.0 References

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